A Data-Centric Comparison of Autoregressive, Energy-Based, and Novel Energy Models for Verifier-Guided Generation

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Planning draft

Abstract

This paper compares three primary model families under one data-centric protocol: an autoregressive language-model baseline, a conventional energy-based model, and a novel free-energy/post-softmax energy model. It also includes diffusion/iterative denoising as a required reference point, because otherwise the comparison artificially excludes the strongest non-autoregressive generative baseline. The contribution is the controlled comparison itself: common splits, common candidate pools, verifier-backed metrics, and matched compute reporting on toy controls, VeriBench/Lean, and an initial MNIST vision track.

1 Thesis

The paper should be publishable even if the novel EBM is only competitive or partially successful. The scientific claim is that architecture debates are underdetermined unless the data, verifier, candidate pool, compute budget, and evaluation protocol are held fixed.

2 Model families

1. **Autoregressive/LLM baseline.** A standard next-token model with teacher forcing, softmax attention, and softmax output head.
2. **Normal EBM baseline.** A conventional energy model trained with established EBM objectives such as contrastive divergence, noise-contrastive estimation, or score-style surrogates where appropriate.
3. **Novel EBM.** The proposed free-energy/post-softmax model from this project.
4. **Diffusion / iterative denoising reference.** A DDPM-style or masked-diffusion-style baseline that pays iterative inference cost rather than local autoregressive normalization.

3 Data

- **Toy controls:** mechanisms with known ground truth.
- **VeriBench/Lean:** primary real-data domain with a hard verifier.
- **MNIST:** first vision smoke test for ordering/global-validity effects.
- **Later vision dataset:** to decide after the MNIST methodology is stable.

4 Metrics

Metric	Purpose
Pass@k	verifier-backed success
cost per verified solution	efficiency under search/retry
success vs length/depth	test error-compounding hypotheses
candidate energy/rank quality	evaluate EBM scoring
matched-compute loss/pass rate	prevent unfair architecture comparisons

5 Experimental protocol

Every run reports the split, model class, parameter count, training compute, inference budget, verifier budget, seeds, and bootstrap intervals where applicable. The data protocol is the paper's main contribution, so leakage boundaries and candidate-pool construction should be described before model results.

6 Expected outcomes

A positive result shows that the novel EBM improves verifier-backed success or cost per verified solution. A useful negative result shows that the data-centric protocol exposes a precise failure mode of EBM training, EBM inference, or AR baselines.